

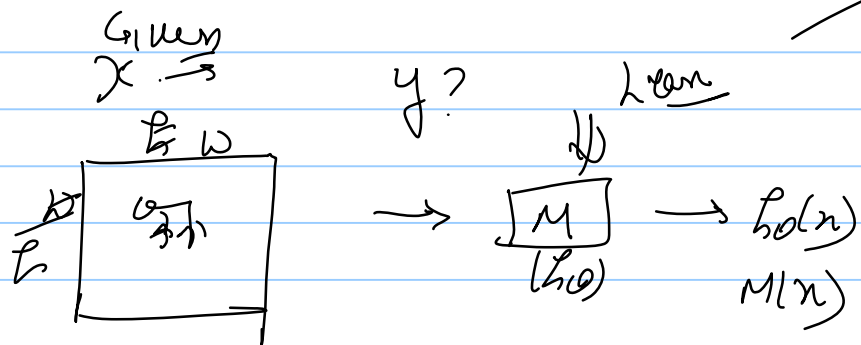
COL774
Aug 1, 2025

Last class -

What is Machine Learning?
AI vs ML

Types of ML

(A) Supervised setting



$\{x^{(i)}, y^{(i)}\}_{i=1}^m$ target variable/output

$\downarrow$ Feature vector

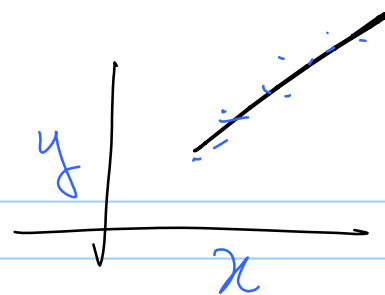
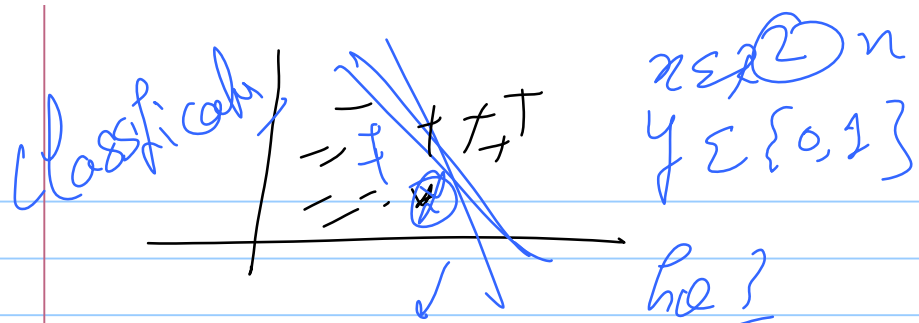
$x^{(i)} \in \mathbb{R}^n$

$\downarrow$ Mix of Discrete & Cont.

$y^{(i)} \in \mathbb{R}$ (regression)

$\hookrightarrow \in \{1, \dots, C\}$ Class, f_i cat

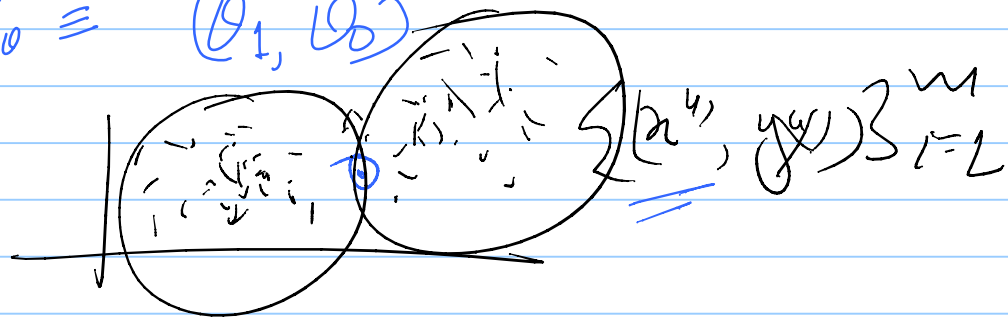
$\in \{0, 1\}$



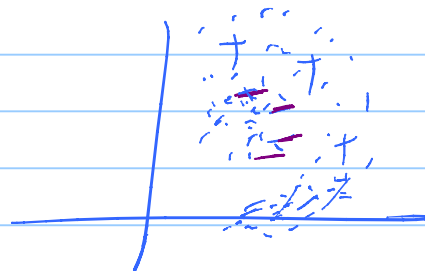
Regression

$h_0 \equiv (w_1, w_0)$

(B)



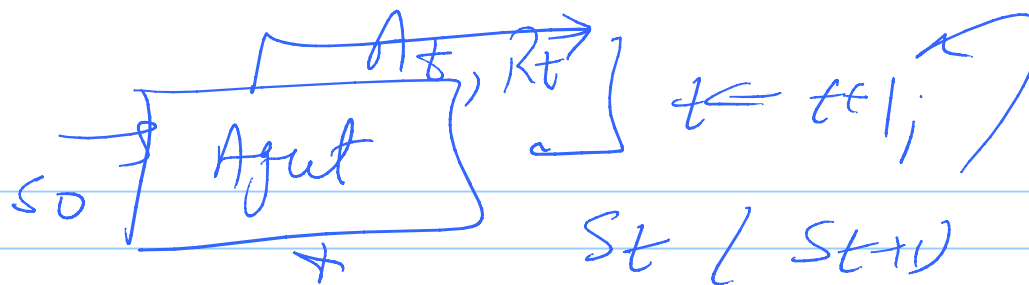
(C) Semi-supervised learning



$\{(x^{(i)}, y^{(i)})\}_{i=1}^m$

$\rightarrow$ missing for a subset of examples

(D) Reinforcement learning



$\beta \in (0, 1)$
 β is discount factor
 $t \geq 0$

$$(\beta < \underline{\underline{1}})$$